ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 4 SET A — SUMMER SEMESTER, 2023-2024

DURATION: 25 MINUTES — FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. A D flip-flop combined with some additional combinational logic is shown in Figure 1. Construct the truth table for the resulting flip-flop, analyzing all possible input combinations. Then, by using process of elimination, determine which standard flip-flop the circuit's behavior most closely resembles. 10 (CO5) (PO3)

   Hint: Consider each case individually and observe how the output changes in response to different inputs and previous states.

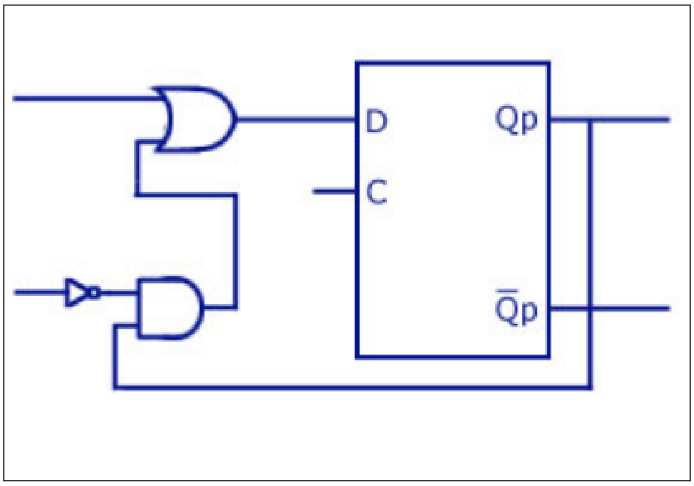

Figure 1: An unknown FlipFlop

Solution:

Step 1: Input Combinations and State Transitions

We examine all combinations of the two inputs (I_1, I_2) and how the output changes relative to the previous state (Q_n).

$$D = I_1 + \overline{I_2} \cdot Q$$

I_1	I_2	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1*
1	1	1	1*

Interpretation:

- When $(I_1, I_2) = (0, 0)$, the output **remains unchanged** regardless of Q_n.
- When $(0, 1)$, the output becomes 0 **Reset**.
- When $(1, 0)$, the output becomes 1 **Set**.
- When $(1, 1)$, the next state is 1*.

Step 2: Compare Behavior with Known Flip-Flops

Improved Deduction Section (LaTeX)

Step 2: Deduction by Elimination of Flip-Flop Types

We now compare the observed behavior with known flip-flop types to identify the correct one.

1. D Flip-Flop

- Has only **one input** (D).
- Characteristic equation: $Q_{n+1} = D$.
- Our circuit has **two inputs**.
- **Conclusion:** Cannot be a D flip-flop.

2. T Flip-Flop

- Has only **one input** (T).
- Behavior:
  - $T = 0 \Rightarrow Q_{n+1} = Q_n$ (no change),
  - $T = 1 \Rightarrow Q_{n+1} = \overline{Q_n}$ (toggle).
- Our circuit does **not toggle** when both inputs are high.
- Also, it has **two inputs**.
- **Conclusion:** Not a T flip-flop.

3. JK Flip-Flop

- Has two inputs: J and K.
- Behavior:
  - $J = 0, K = 0 \Rightarrow Q_{n+1} = Q_n$,
  - $J = 0, K = 1 \Rightarrow Q_{n+1} = 0$,
  - $J = 1, K = 0 \Rightarrow Q_{n+1} = 1$,
  - $J = 1, K = 1 \Rightarrow Q_{n+1} = \overline{Q_n}$ (toggle).
- In our case, for input (1, 1), the output **does not toggle**; it remains at 1.
- **Conclusion:** Behavior does not match JK flip-flop **Not a JK flip-flop**.

4. SR Flip-Flop

- Has two inputs: S (Set) and R (Reset).
- Truth table:

S	R	Q_{n+1}
0	0	Q_n (No change)
0	1	0 (Reset)
1	0	1 (Set)
1	1	Invalid

- Matches observed behavior:
  - (0, 0) No change
  - (0, 1) Reset (Q becomes 0)
  - (1, 0) Set (Q becomes 1)
  - (1, 1) Invalid / undefined
- **Conclusion:** The behavior matches the SR flip-flop more than the others if we consider 11 condition to be invalid/don't care.

Therefore, the flip-flop is an SR flip-flop.
Any other logical deduction regarding the 11 condition will be accepted, provided it is supported by valid justification. .
Rubrics:

- 2 points each (times 4) for correctly identifying each case.
- 2 points for correctly identifying the resultant flip flop's behavior with proper justification.

2. Design a 5-bit Parallel-In Serial-Out (PISO) shift register **(shifting right to left)** with a control line labeled SHIFT\$\overline{\text{LOAD}}$. **Explain the working procedure of the circuit in different modes.** 5 + 5 (CO5) (PO3)

Solution:

Working Procedure and Mode Justification

The 5-bit Parallel-In Serial-Out (PISO) shift register operates in two distinct modes based on the control line labeled SHIFT$\backslash\overline{\text{LOAD}}$. The behavior of the circuit in each mode is as follows:

- **Mode 1: Load Mode**
  When the control line is **low** ($\overline{\text{LOAD}} = 0$), the circuit enters **parallel load mode**. In this mode:
  - The parallel input lines (D_4 to D_0) are enabled.
  - Each flip-flop in the register captures its corresponding input bit on the rising edge of the clock.
  - No shifting occurs during this mode.
- **Mode 2: Shift Mode**
  When the control line is **high** (SHIFT = 1), the circuit enters **serial shift mode**. In this mode:
  - The parallel inputs are disabled.
  - On each clock pulse, the data stored in the flip-flops shifts one bit to the left (i.e., from right to left).
  - The leftmost bit (e.g., D_4) is shifted out to the serial output line.
  - A logic 0 or a new serial input (if available) may be shifted into the rightmost flip-flop.

Summary: The control line effectively determines whether the register captures parallel input (load mode) or shifts its stored data (shift mode). By controlling the timing of load and shift signals, a 5-bit word can be loaded and then shifted out serially, bit by bit, starting from the leftmost bit (MSB) toward the rightmost bit (LSB).
Rubrics:

- 1 point for correctly showing the modes(2).
- 1 point for correctly using D flip flop (or any other with proper justification) (5).
- 2 point for correctly using the AND Gates (10).
- 1 points for correctly using the OR Gates (5).
- 2.5 points for correctly explaining the working mechanism of each mode (times 2).